Phase 3 (part 3): Centrality and Matrix Methods

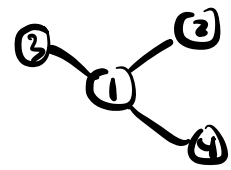
1.] Consider the unweighted, undirected network on nodes $\{1, 2, 3, 4\}$ with edges

↳ bemerick_p3ws3_1.py

$$\{(1, 2), (1, 3), (1, 4)\}.$$

a.) Compute the degree of each node.

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \qquad \begin{aligned} \deg(1) &= 3 \\ \deg(2) = \deg(3) &= \deg(4) = 1 \end{aligned}$$



b.) The vector $\boldsymbol{x}$ is an eigenvector of A if it is nonzero and satisfies the equation

$$A\boldsymbol{x} = \lambda \boldsymbol{x}$$

for some scalar λ. Write the system of equations relating x_1, x_2, x_3, x_4 and solve for possible values of λ.

- $\lambda_1 = 0$ and $\lambda_2 = 0$ make up 2 of the 4 eigenvalues.
- Write out the system:

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_2 + x_3 + x_4 \\ x_1 \\ x_1 \\ x_1 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda x_2 \\ \lambda x_3 \\ \lambda x_4 \end{bmatrix} \rightarrow \begin{cases} x_2 + x_3 + x_4 = \lambda x_1 \\ x_1 = \lambda x_2 \\ x_1 = \lambda x_3 \\ x_1 = \lambda x_4 \end{cases} \rightarrow \begin{cases} x_2 + x_3 + x_4 = \lambda x_1 \\ x_2 = \frac{1}{\lambda} x_1 \\ x_3 = \frac{1}{\lambda} x_1 \\ x_4 = \frac{1}{\lambda} x_1 \end{cases}$$

$$\hookrightarrow \frac{1}{\lambda} x_1 + \frac{1}{\lambda} x_1 + \frac{1}{\lambda} x_1 = \lambda x_1 \rightarrow \frac{3}{\lambda} x_1 = \lambda x_1 \rightarrow 3x_1 = \lambda^2 x_1 \rightarrow \boxed{\begin{aligned} \lambda_3 &= -\sqrt{3} \\ \lambda_4 &= \sqrt{3} \end{aligned}}$$

c.) Find the eigenvector corresponding to the highest positive eigenvalue.

$$\underline{\lambda_4 = \sqrt{3}}: \quad \begin{aligned} x_2 + x_3 + x_4 &= \sqrt{3}\, x_1 \\ x_2 &= \tfrac{1}{\sqrt{3}} x_1 \\ x_3 &= \tfrac{1}{\sqrt{3}} x_1 \\ x_4 &= \tfrac{1}{\sqrt{3}} x_1 \end{aligned} \rightarrow \begin{cases} \frac{3}{\sqrt{3}} x_1 = \sqrt{3}\, x_1 \\ x_2 = \frac{1}{\sqrt{3}} x_1 \\ x_3 = \frac{1}{\sqrt{3}} x_1 \\ x_4 = \frac{1}{\sqrt{3}} x_1 \end{cases} \rightarrow \text{Pick } x_1 = \sqrt{3} \rightarrow \boxed{\vec{v}_4 = \begin{bmatrix} \sqrt{3} \\ 1 \\ 1 \\ 1 \end{bmatrix}}$$

d.) Which node has the highest centrality? Does this match your intuition?

$$|\vec{v}_4| = \sqrt{3+1+1+1} = \sqrt{6}$$

unit eigenvector

$$\hookrightarrow \frac{\vec{v}_4}{|\vec{v}_4|} = \vec{u}_4 = \begin{bmatrix} \sqrt{3}/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} = \begin{bmatrix} \sqrt{1/2} \\ \sqrt{1/6} \\ \sqrt{1/6} \\ \sqrt{1/6} \end{bmatrix}$$

Node 1 has the highest eigenvector centrality at $\sqrt{1/2}$

2.] Consider the weighted graph with edges:

$$(A, B, 3), (B, C, 2), (A, D, 4), (B, E, 5), (C, F, 1), (D, E, 2), (E, F, 3), (B, D, 2).$$

a.) Build this graph in Python using `networkx`. ⇝ bemerick_p3ws3_2.py

b.) Compute the strength of each node. Which node has the largest strength?

$$\begin{pmatrix} 0 & 3 & 0 & 4 & 0 & 0 \\ 3 & 0 & 2 & 2 & 5 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 \\ 4 & 2 & 0 & 0 & 2 & 0 \\ 0 & 5 & 0 & 2 & 0 & 3 \\ 0 & 0 & 1 & 0 & 3 & 0 \end{pmatrix} \rightarrow \text{Strength} = \begin{pmatrix} 7 \\ 12 \\ 3 \\ 8 \\ 10 \\ 4 \end{pmatrix}$$

Node B has the highest strength.

c.) Compute the closeness centrality, betweenness centrality, and eigenvector centrality. List the nodes in order from largest to smallest for each centrality.

Using Python directly, we have

	B	D	E	C	A	F
closeness:	0.83	0.71	0.71	0.63	0.56	0.56

	B	E	C	D	F	A
betweenness:	0.37	0.22	0.08	0.08	0.05	0.00

	B	D	E	A	C	F
eigenvector:	0.55	0.47	0.45	0.36	0.28	0.26

d.) Do all the centrality rankings agree? If not, why might different centralities disagree?

The all agree that B has the highest centrality but the order of rankings differs for each definition of centrality. They differ because each method uses a different definition of and computation for centrality.